NETWORK / SOC ENGINEER L1 · INTERVIEW & FIELD GUIDE · PART 2 OF 2

Switching, Routing, TCP/IP & Firewall Troubleshooting

Theory · Configuration · Packet flow · Wireshark captures explained · Scenario questions · Customer-network troubleshooting

OSI & TCP/IP | ARP / ICMP | TCP / UDP | DNS / DHCP | Subnetting / NAT | VLAN / STP / EtherChannel | OSPF / BGP / HSRP | Palo Alto / IPSec / GlobalProtect | Wireshark

ABOUT THE WIRESHARK SCREENSHOTS

The capture images in this guide are Wireshark-style renderings of realistic example packets (field names, values and colours follow what Wireshark shows). They are for learning to read captures. They are not live screenshots from a real network, so IPs and MACs are lab examples. Open the same traffic in your own lab (GNS3 / EVE-NG / Packet Tracer + Wireshark) and compare.

Contents: Part 2 of 2

9	Firewall: Palo Alto flow, NAT, IPSec, GlobalProtect, Panorama
10	Wireshark field guide and filters
11	How routers communicate: hop-by-hop worked example
12	Troubleshooting a customer network (architecture + scenarios)
13	Rapid-fire interview Q&A
14	Last-day revision plan and cheat sheet

Part 1 contains: 1. OSI, TCP/IP and end-to-end packet flow; 2. Ethernet, MAC and ARP; 3. IP, ICMP, ping and traceroute; 4. TCP and UDP; 5. DNS and DHCP; 6. Subnetting and NAT/PAT; 7. Switching: VLAN, trunk, STP, EtherChannel, switch security; 8. Routing: static, OSPF, BGP, HSRP.

HOW TO USE THIS GUIDE

Order of study: chapters 1, 4, 2, 7 (OSI, TCP, ARP, switching) first, as you asked, then 8, 5, 3, 6, then Part 2 (firewall, Wireshark, router communication, customer troubleshooting, Q&A). Every chapter ends with scenario questions so you practise answering aloud.

9. Firewall: Palo Alto flow, NAT, IPSec, GlobalProtect, Panorama

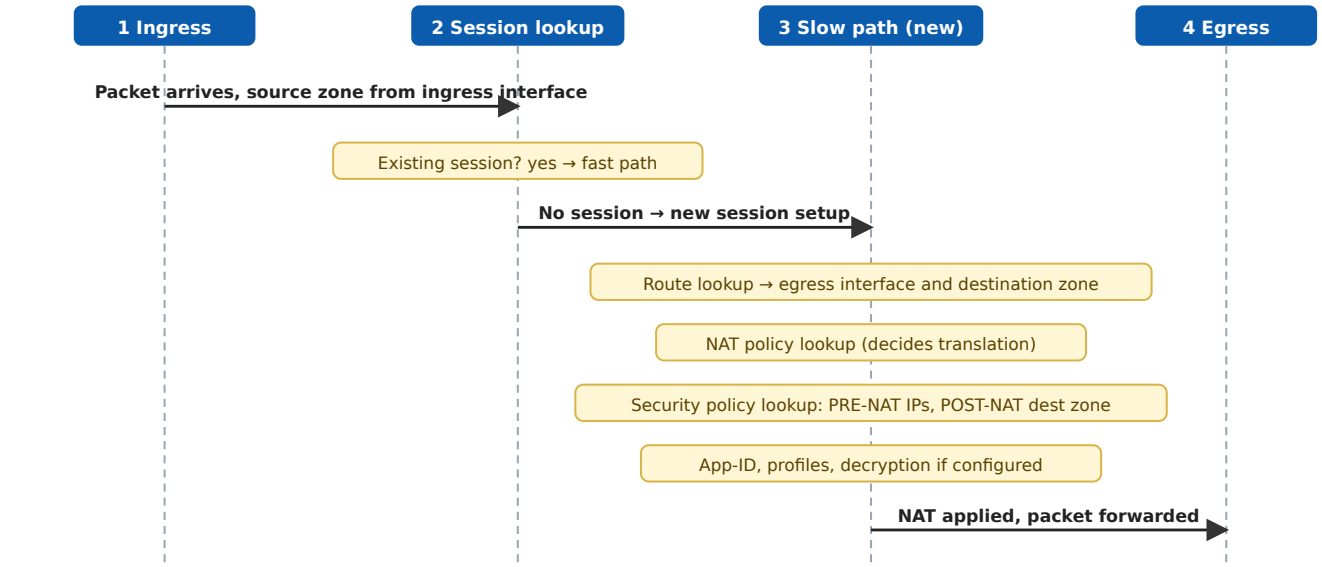
Your strongest area: be precise here

9.1 Core concepts

- **Zones** group interfaces (trust, untrust, dmz). Policy is zone-based. Interface types: Layer3, Layer2, Virtual Wire, Tap. **Virtual router** holds routes.
- **Security policy:** top-down, first match wins. Match on source/destination zone, address, user, **application (App-ID)**, service, URL category. Defaults: *intrazone-default = allow*, *interzone-default = deny* (not logged unless enabled).
- **Content-ID / profiles:** antivirus, anti-spyware, vulnerability, URL filtering, file blocking, WildFire. **User-ID** maps IP to user.
- **Session:** first packet goes through the full path (slow path); later packets use the session (fast path). Default timeouts: TCP 3600 s, UDP 30 s, ICMP 6 s.

First-packet processing order (simplified)

Policy uses the original addresses, but the destination zone is the one after NAT/route



9.2 Troubleshooting method: user cannot reach an application through the firewall

1. **Logs:** Monitor > Traffic. Filter by source/destination. Note rule, action, application, *session end reason* (policy-deny, aged-out, tcp-rst-from-client/server, threat, decrypt error). No log at all = traffic not arriving (routing/upstream) or implicit deny not logged.
2. **Zones/routing:** is traffic arriving on the expected interface/zone? `show routing route destination 10.1.1.50`, `test routing fib-lookup virtual-router default ip 10.1.1.50`.
3. **Policy:** `test security-policy-match from trust to untrust source 192.168.10.10 destination 10.1.1.50 destination-port 443 protocol 6`. Check rule order and shadowing.
4. **NAT:** `test nat-policy-match ...` and `show session id N` (shows c2s and s2c flows with NAT).
5. **Session:** `show session all filter source 192.168.10.10 destination 10.1.1.50`.
6. **Counters:** `show counter global filter delta yes severity drop` (e.g. flow_policy_deny, flow_tcp_non_syn, flow_fwd_l3_nortr, flow_session_notfound).
7. **Packet capture:** GUI Monitor > Packet Capture with stages receive / firewall / transmit / drop. Compare what arrived with what left.

```
> show session all filter source 192.168.10.10 destination 10.1.1.50 destination-port 443
ID      Application  State   Type Flag  Src[Sport]/Zone/Proto (translated IP[Port])     Dst[Dport]/Zone
1234    ssl          ACTIVE  FLOW  NS   192.168.10.10[51514]/trust/6 (203.0.113.2[40001])  10.1.1.50[443]/untrust
> show session id 1234
> test security-policy-match from trust to untrust source 192.168.10.10 destination 10.1.1.50 destination-port 443 protocol 6
> test nat-policy-match from trust to untrust source 192.168.10.10 destination 10.1.1.50 destination-port 443 protocol 6
> show counter global filter delta yes severity drop
> debug dataplane packet-diag set filter match source 192.168.10.10 destination 10.1.1.50
> debug dataplane packet-diag set filter on
> debug dataplane packet-diag set capture stage receive file rx.pcap
> debug dataplane packet-diag set capture on      ! reproduce, then: set capture off
> view-pcap filter-pcap rx.pcap
> debug dataplane packet-diag clear all
```

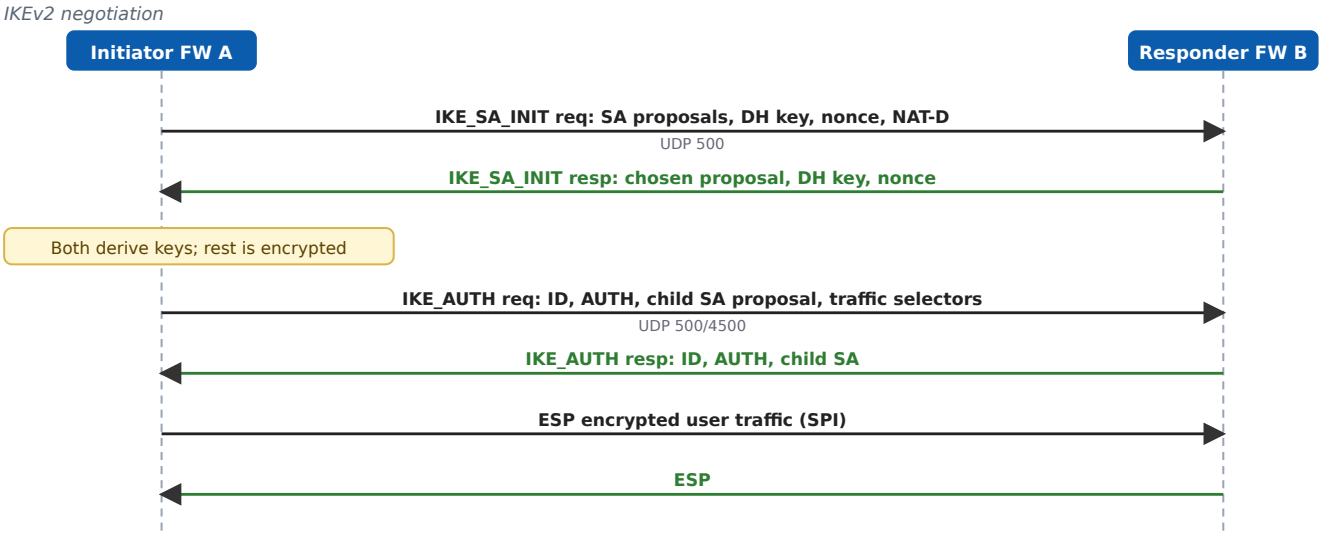
Log / symptom	Meaning	Action
Session end: policy-deny / action deny	Matched a deny or the interzone default	Fix rule order/match; check zone, App-ID, service
Application: incomplete	TCP handshake did not complete, or no data after it	Server not responding, return path, asymmetric routing
Application: insufficient-data / unknown-tcp	Not enough data to identify or custom app	Allow by service/port temporarily, create custom App-ID
Session end: aged-out, no reply	Silent drop upstream or no return traffic	Capture on both sides, check routing and server
flow_tcp_non_syn drops	Mid-stream packet without a session: asymmetric path or session expired	Make routing symmetric, check HA, timeout
flow_fwd_l3_nortr	No route	Add route in the right virtual router
Works for IP, fails for FQDN	DNS or FQDN object not resolved	Check DNS proxy / object refresh
decrypt-cert-validation / decrypt-unsupport-param	SSL decryption issues	Add no-decrypt exception, trust the CA, check cipher

9.3 NAT on the firewall

- Source NAT (dynamic IP and port) for internet access, destination NAT for published servers, static NAT for one-to-one.
- NAT rules are evaluated top-down, separate from security policy. Zones in the NAT rule are original source zone and **original** destination zone.
- Published server needs: NAT rule (public to private) **and** security rule with **public (pre-NAT) destination IP**, destination zone = zone of the private server.
- Server's default route must point back through the firewall (otherwise asymmetric).

9.4 IPSec VPN

	Phase 1 (IKE SA)	Phase 2 (IPSec / Child SA)
Purpose	Authenticate peers, create secure control channel	Create tunnels that carry user traffic
Parameters	Auth (PSK/cert), encryption, hash/PRF, DH group, lifetime, peer ID	Encryption, authentication, PFS/DH group, lifetime, proxy IDs / traffic selectors
Ports/protocol	UDP 500; UDP 4500 if NAT-T	ESP (IP protocol 50) or ESP in UDP 4500
IKEv1	Main mode (6 msgs) or aggressive (3)	Quick mode (3 msgs)
IKEv2	IKE_SA_INIT + IKE_AUTH (4 msgs)	Child SA is created inside IKE_AUTH; CREATE_CHILD_SA for rekeys



Wireshark: IKEv2 and ESP

ike.pcapng - Wireshark

Apply a display filter ... `isakmp || esp || udp.port==500 || udp.port==4500`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	198.51.100.10	203.0.113.2	ISAKMP	450	IKE_SA_INIT MID=00 Initiator Request
2	0.031000	203.0.113.2	198.51.100.10	ISAKMP	410	IKE_SA_INIT MID=00 Responder Response
3	0.060000	198.51.100.10	203.0.113.2	ISAKMP	322	IKE_AUTH MID=01 Initiator Request
4	0.095000	203.0.113.2	198.51.100.10	ISAKMP	286	IKE_AUTH MID=01 Responder Response
5	1.000000	198.51.100.10	203.0.113.2	ESP	142	ESP (SPI=0xc3a1f0e2)
6	1.021000	203.0.113.2	198.51.100.10	ESP	142	ESP (SPI=0x1b9d44aa)
7	60.000000	198.51.100.10	203.0.113.2	ISAKMP	450	IKE_SA_INIT MID=00 Initiator Request
8	60.030000	203.0.113.2	198.51.100.10	ISAKMP	90	IKE_SA_INIT MID=00 Responder Response NOTIFY: NO_PROPOSAL_CHOSEN

```
▸ Internet Protocol Version 4, Src: 198.51.100.10, Dst: 203.0.113.2
▾ User Datagram Protocol, Src Port: 500, Dst Port: 500 (1)
▾ Internet Security Association and Key Management Protocol
      Initiator SPI: 8c4d2e1a9b7f6a31 (2)
      Responder SPI: 0000000000000000
      Next payload: Security Association (33)
      Version: 2.0
      Exchange type: IKE_SA_INIT (34) (3)
      Flags: 0x08 (Initiator, No higher version, Request) (4)
      Message ID: 0x00000000
      Length: 408
    ▾ Payload: Security Association (33) (5)
          Proposal: ENCR_AES_CBC (256), PRF_HMAC_SHA2_256, AUTH_HMAC_SHA2_256_128, DH group 14 (2048-bit MODP)
    ▸ Payload: Key Exchange (34)
    ▸ Payload: Nonce (40)
    ▾ Payload: Notify (41) - NAT_DETECTION_SOURCE_IP (16388) (6)
    ▸ Payload: Notify (41) - NAT_DETECTION_DESTINATION_IP (16389)
```

#	What it denotes
1	IKE on UDP 500. If a NAT device is detected in the path, later messages move to UDP 4500 (NAT-T) and ESP is wrapped in UDP.
2	Initiator SPI identifies the IKE SA. Responder SPI is zero in the first message and filled in by the reply.
3	Exchange types: IKE_SA_INIT 34, IKE_AUTH 35, CREATE_CHILD_SA 36, INFORMATIONAL 37.
4	Flag bits: Initiator (0x08), Response (0x20). Only requests with no reply = peer unreachable or UDP 500 blocked.
5	**Proposal** lists encryption, integrity/PRF, DH group. Must overlap with the peer's config. Packet 8: **NO_PROPOSAL_CHOSEN** (notify 14) = no common proposal in Phase 1.
6	NAT-Detection payloads: each side hashes IP/port. A mismatch tells both there is NAT, so they switch to NAT-T (4500). Packets 5-6 show ESP with a different SPI each direction; if you see ESP only one way, the return path is broken.

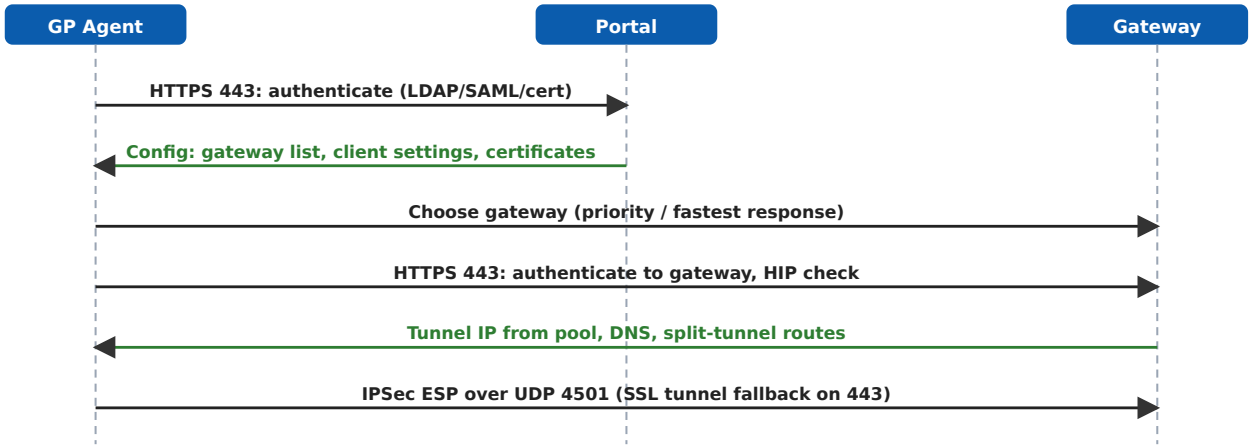
Figure 9.1: Tunnel bring-up (1-4), encrypted traffic (5-6), and a failed negotiation (7-8). ESP payload cannot be read without keys.

VPN problem	Check
Phase 1 down, no response from peer	Peer IP/route, UDP 500/4500 allowed upstream, interface/zone allows IKE (management profile or policy), peer configured. `show vpn ike-sa` , `less mp-log ikemgr.log` .
NO_PROPOSAL_CHOSEN / proposal mismatch	Align encryption, hash, DH group, IKE version between both ends (crypto profile).
Authentication failed	PSK mismatch, wrong local/peer ID, certificate issue.

Phase 1 up, Phase 2 fails	Proxy IDs / traffic selectors mismatch (Cisco ACL vs PAN proxy ID), PFS mismatch, wrong IPSec crypto profile.
Tunnel up, no traffic	Route to remote subnet via tunnel interface, security policy VPN zone, NAT exemption, overlapping subnets, tunnel monitor. `show vpn flow` (encap/decap counters): encap increasing but decap zero = remote not returning.
Tunnel drops periodically	Lifetime/rekey mismatch, dead peer detection, ISP flapping, NAT timeout (enable NAT keepalive).
Big transfers fail	MTU/MSS: adjust tunnel interface MTU / TCP MSS clamp.

9.5 GlobalProtect (SSL/IPSec remote access VPN)

GlobalProtect connection flow



GlobalProtect issue	Likely cause / check
Cannot connect to portal	DNS of portal FQDN, TCP 443 reachable (Wireshark SYN, no SYN-ACK), portal interface and policy, certificate trust (client does not trust the portal cert or chain).
Authentication fails	Auth profile (LDAP bind, SAML metadata/clock skew, RADIUS secret), user not in allowed list, certificate profile. Check system logs and `authd.log`.
Connected but no access to resources	Split-tunnel routes/access routes, security policy from GP zone to internal zone, NAT for tunnel pool, return route to pool on internal network.
Internal names do not resolve	DNS server/suffix not set in gateway config, DNS policy.
Frequently falls back to SSL / slow	UDP 4501 blocked on the user's network, so ESP-in-UDP fails, tunnel falls to SSL.
HIP check failing	HIP profile and object mismatch (patch/AV state); check HIP report in logs.
Check live users	`show global-protect-gateway current-user`; agent logs `PanGPS.log`, `PanGPA.log` on client.

9.6 Panorama and HA

- **Panorama** centrally manages firewalls. **Device groups** hold policies and objects (shared / pre-rules / post-rules). **Templates and template stacks** hold network and device settings (interfaces, zones, routing, profiles). **Commit** saves to Panorama, then **Push to Devices** applies it. Firewalls connect to Panorama on TCP 3978.
- Common issues: device shows *disconnected* (management reachability, TCP 3978, serial/auth key), *out of sync* (push needed or local override), commit failure from validation errors (missing zone/interface in template), template variable mismatch. Check `show panorama-status` (firewall) and `show devices connected` (Panorama).
- **High availability:** active/passive pair. HA1 control link and heartbeat, HA2 session/state sync, optional backup links. Failover triggers: link/path monitoring, heartbeat loss, manual suspend. Preemption optional. Verify `show high-availability state` and config sync status. Mismatched versions, licenses, or HA modes block sync.

> **INTERVIEW ANGLE**
> **"Walk me through a VPN tunnel down ticket."** Confirm which phase fails with *show vpn ike-sa* and logs (ikemgr.log, system log). Phase 1: reachability on UDP 500/4500, proposal, PSK, IDs. Phase 2: proxy IDs/selectors, PFS, crypto profile. Then data plane: routes, policy, NAT, encap/decap counters, MTU. Capture IKE if logs are unclear. Escalate with the logs, config of both sides, and timestamps.

10. Wireshark field guide and filters

Capturing, filtering and reading quickly

10.1 Capture workflow

1. Capture at the right place: client, server, firewall interface or SPAN port. Capture on both sides to localise loss.
2. Use **capture filters** (BPF) to limit what is recorded, e.g. `host 10.1.1.50 and port 443`. Use **display filters** to analyse afterwards.
3. Keep captures short, reproduce the issue once, note time of the failure.
4. Useful views: Statistics > Conversations, Protocol Hierarchy, I/O Graph, Analyze > Expert Information, right-click > Follow > TCP Stream.

Display filter	Use
`ip.addr == 10.1.1.50`, `ip.src ==`, `ip.dst ==`	Host traffic
`tcp.port == 443`, `udp.port == 53`	By port
`tcp.flags.syn==1 && tcp.flags.ack==0`	New connection attempts
`tcp.flags.reset==1`	Resets
`tcp.analysis.retransmission || tcp.analysis.duplicate_ack || tcp.analysis.zero_window`	TCP problems
`tcp.analysis.flags`	All TCP expert flags
`dns`, `dns.flags.rcode != 0`	DNS, DNS errors
`dhcp`, `bootp.option.dhcp == 1`	DHCP, Discover
`arp`, `arp.duplicate-address-detected`	ARP, duplicate IP
`icmp.type == 3 || icmp.type == 11`	ICMP errors
`stp`, `ospf`, `bgp`, `hsrp`, `isakmp`, `esp`	Protocols
`vlan.id == 20`	VLAN
`http.request || tls.handshake.type == 1`	HTTP requests / TLS Client Hello
`frame.time_delta > 1`	Gaps longer than 1 s
`tcp.stream eq 5`	One conversation

Capture filter (BPF)	Use
`host 10.1.1.50`	One host
`net 192.168.10.0/24`	Subnet
`port 53`, `tcp port 443`	Ports
`not arp and not broadcast`	Remove noise
`udp port 500 or udp port 4500`	IPSec negotiation

10.2 Reading a capture: 5-step approach

1. **Protocol Hierarchy / Conversations**: who talks most, anything unexpected.
2. **Expert Information**: retransmissions, resets, malformed packets, duplicate IPs.
3. Pick the failing conversation (`tcp.stream`) and read the handshake.
4. Check **time deltas** (`tcp.time_delta`, `dns.time`) to separate network delay from server delay.
5. Compare client-side and server-side captures: where does the packet disappear or change (NAT, TTL, MTU, flags)?

Colour in Wireshark (default)	Typically
Light purple	TCP
Light blue	UDP / DNS / DHCP
Pink	ICMP
Pale yellow/peach	ARP / routing protocols
Green	HTTP
Black with red text	TCP problems (retransmission, dup ACK, out-of-order)

Red with white text	TCP RST

11. How routers communicate: hop-by-hop worked example

What really changes in a packet as it crosses the network

Topology: PC `192.168.10.10` → R1 (LAN `192.168.10.1`, WAN `10.0.0.1`) → R2 (WAN `10.0.0.2`, LAN `172.16.1.1`) → Server `172.16.1.50`. PC pings the server for the first time (all ARP caches empty).

Step	What happens
1	PC: destination is not in its subnet (mask /24), so it needs the gateway MAC. It sends **ARP request** (broadcast) for 192.168.10.1. R1 replies with its LAN MAC.
2	PC sends the ICMP echo: **Eth dst = R1 LAN MAC**, IP dst = 172.16.1.50, TTL 128.
3	R1: strips the L2 header, looks up 172.16.1.50 in the routing table (OSPF/static via 10.0.0.2), decrements TTL to 127, recomputes IP checksum. ARPs for 10.0.0.2 if needed.
4	R1 builds a **new Ethernet header**: src = R1 WAN MAC, dst = R2 WAN MAC. IPs unchanged.
5	R2: same process. Destination is directly connected, so it ARPs for 172.16.1.50. TTL 126. New frame: src = R2 LAN MAC, dst = server MAC.
6	Server replies. Its gateway is R2, so the reply goes server → R2 → R1 → PC. The return path needs its own routes: **both directions must be routed.**

Segment	Eth src	Eth dst	IP src	IP dst	TTL
PC → R1	PC MAC	R1 LAN MAC	192.168.10.10	172.16.1.50	128
R1 → R2	R1 WAN MAC	R2 WAN MAC	192.168.10.10	172.16.1.50	127
R2 → Server	R2 LAN MAC	Server MAC	192.168.10.10	172.16.1.50	126

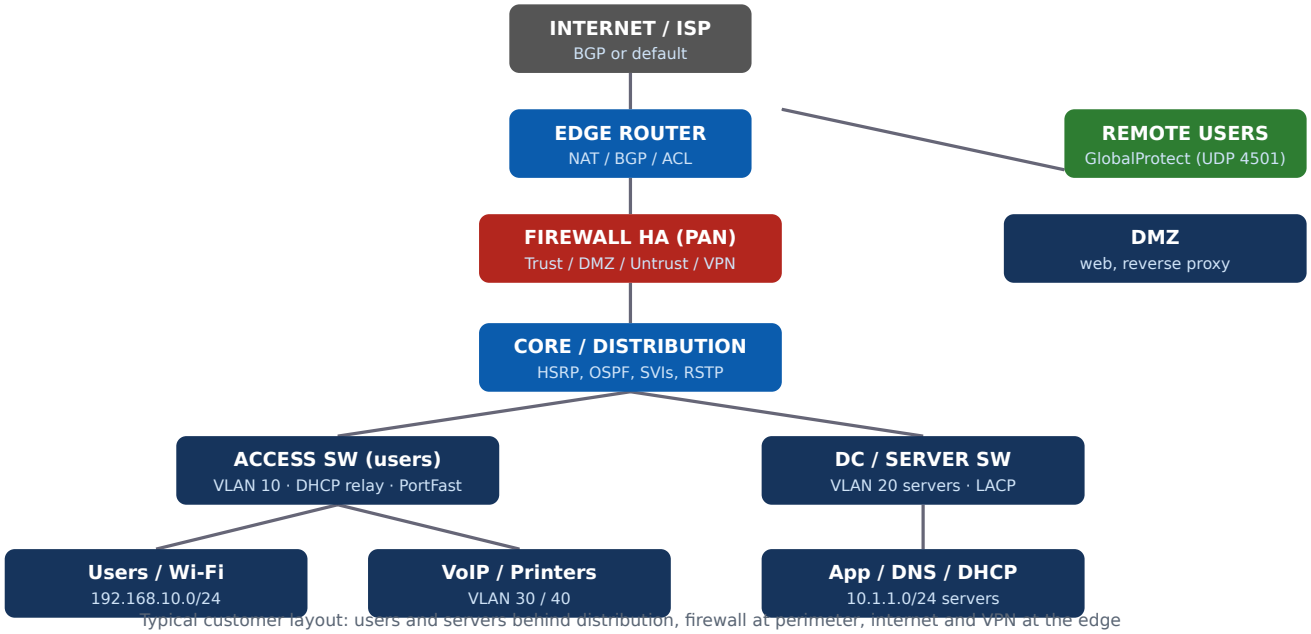
If this fails...	Symptom	Cause
ARP for gateway	PC shows incomplete ARP entry	VLAN/port/gateway down
R1 has no route	ICMP net unreachable from R1 (type 3/0)	Missing route/OSPF
R1 cannot ARP next hop	Host unreachable (3/1) or timeout	Link/VLAN problem on WAN segment
R2 has no return route	Echo reaches server, reply never returns; timeout	Missing route to 192.168.10.0/24 on server side
TTL reaches 0	Time exceeded (11/0) from the looping router	Routing loop
Firewall in path	Timeout with no ICMP error	Policy, asymmetry

INTERVIEW ANGLE

"What changes and what stays the same as a packet crosses routers?" MACs change every hop, TTL drops by one, header checksum is recomputed; IPs and ports stay unless NAT is applied; payload is unchanged.

12. Troubleshooting a customer network

Architecture-first method and scenarios



Typical customer layout: users and servers behind distribution, firewall at perimeter, internet and VPN at the edge

Reference architecture used in the scenarios below. Always ask the customer for a diagram, device list, IP plan and what changed recently.

12.1 Universal approach

1. **Define the problem:** who, what, since when, how many users, what changed? Get source/destination IP, port, application and exact error.
2. **Scope:** one user, one VLAN, one site, all? One destination or all? This maps to access switch, distribution/gateway, firewall, or WAN/ISP.
3. **Follow the path** from source to destination on the diagram, layer by layer: link → VLAN → gateway → routing → firewall/NAT → server → application.
4. **Test both directions** and from both ends (extended ping with source, traceroute, port test).
5. **Collect evidence**: logs, counters, packet capture, config diff, then change one thing at a time.
6. **Escalate/document** with timeline, scope, tests run, evidence, and impact.

Scope	Think first
One user	NIC/cable, VLAN, IP/DHCP, port security, local firewall, user policy
One VLAN / floor	Access switch, uplink trunk, STP, gateway SVI/HSRP, DHCP relay
All users to the internet	Firewall, NAT, edge router, ISP link, DNS, default route
One app only	Firewall policy/App-ID, server, DNS, load balancer, certificate
Remote users only	GlobalProtect/VPN, portal cert, auth, pool routing
Intermittent	Loop/flap, duplicate IP, HA failover, errors on link, asymmetric path, session timeout

12.2 Scenarios

S1: ONE USER HAS NO NETWORK

Link up, but 169.254.x.x. Check VLAN on port (*show vlan brief*), port-security/err-disable, DHCP: capture shows Discover with no Offer? Check helper-address, scope, DHCP snooping trust on uplink. Fix and renew with *ipconfig /renew*.

S2: WHOLE FLOOR OFFLINE

Access switch uplink down or STP/trunk issue. Check uplink interface status and logs, SFP light levels, port-channel state, STP topology changes, native/allowed VLAN changes. Use redundant path if available. Escalate L2 hardware if the port or fibre is faulty.

S3: USERS CAN PING GATEWAY BUT NOT THE INTERNET
Trace upstream: *traceroute 8.8.8.8* to find the last working hop. Firewall: traffic log for user IP, NAT, route to internet, edge router, ISP status. Test DNS separately (*nslookup*).

S4: INTERNET WORKS, INTERNAL APP DOES NOT
Ping server IP, test TCP port. If SYN gets no reply: firewall policy/route/server. If RST: service down or reject. Check DNS name, certificate, server health, load balancer.

S5: APPLICATION SLOW
Complaint: 'network is slow' for one app. Measure: ping latency/loss, capture handshake RTT, server response delay (*tcp.time_delta*), retransmissions, interface errors/utilisation on path, duplex mismatch, QoS drops, firewall CPU/session limits, DNS delay.

S6: VPN SITE-TO-SITE DOWN
Branch cannot reach HQ. show vpn ike-sa/ipsec-sa; Phase 1 vs Phase 2 logs; ISP/peer IP change; PSK/proposal changes; routes and policy; encap/decap counters; MTU.

S7: GLOBALPROTECT USERS CANNOT CONNECT
Portal reachable (443)? Certificate valid? Auth backend working? Logs in firewall, user count vs license, gateway interface, UDP 4501 and SSL fallback, after-connect routes and policy.

S8: INTERMITTENT DROPS EVERY FEW MINUTES
Check: duplicate IP/MAC flap logs, STP TCN, HSRP flapping, HA failover logs, interface flaps/CRC, session timeouts, DHCP lease conflicts, ISP link flaps. Correlate timestamps across devices.

S9: AFTER HSRP/HA FAILOVER SOME APPS FAIL
Existing sessions lost (non-synced), asymmetric path, ARP/MAC not updated, tracking misconfigured, secondary device missing route or policy. Compare configs, check sync state.

S10: SERVER UNREACHABLE FROM OUTSIDE
Published web server. Check DNS public IP, NAT rule, security policy (pre-NAT dest, correct zone), server gateway returns via firewall, listening port, upstream ACL, ISP route.

S11: OSPF/BGP NEIGHBOUR DOWN
OSPF: area, timers, mask, MTU, auth, passive. BGP: TCP 179 reachability, AS/neighbour IP, auth, capabilities, prefix filters. Look at interface state and recent change first.

S12: BROADCAST STORM
Network unusable, CPU 100%. Identify looping ports by broadcast counters/MAC flaps, remove loop, verify STP root/BPDU guard, then restore services. Document root cause (unmanaged switch, patch loop).

12.3 Escalation note template

```
Ticket: INC-xxxxx   Priority: P2   Time detected: 14:05 IST
Impact: 40 users on floor 3 cannot reach ERP (10.1.1.50:443). Others unaffected.
Findings: Gateway reachable; SYN sent, no SYN-ACK; firewall traffic log shows deny by rule 'interzone-default'.
Tests done: ping/traceroute (path OK to firewall), test security-policy-match, capture on client.
Recent change: new VLAN 31 added today (no policy for zone USERS-31).
Action taken / requested: need L2 firewall team to add rule or confirm zone assignment. Evidence attached (pcap, logs).
```

13. Rapid-fire interview Q&A

Short answers you can say aloud

Question	Short answer
What is the difference between a hub, switch and router?	Hub repeats to all ports (L1). Switch forwards by MAC per port (L2). Router forwards by IP between networks (L3).
Collision domain vs broadcast domain?	Each switch port is a collision domain; a VLAN/subnet is a broadcast domain. Routers and VLANs split broadcast domains.
What happens when a switch receives a frame with unknown destination MAC?	Floods it out all ports in the VLAN except the ingress port.
Access vs trunk port?	Access carries one untagged VLAN; trunk carries multiple VLANs tagged with 802.1Q (except native).
What is native VLAN?	Untagged VLAN on a trunk; must match both ends.
How is the STP root bridge chosen?	Lowest bridge ID = priority + VLAN ID + MAC.
PortFast and BPDU Guard?	PortFast skips listening/learning on edge ports. BPDU Guard err-disables the port if it receives a BPDU.
RSTP vs STP?	RSTP converges in seconds with proposal/agreement and alternate/backup roles.
What is an EtherChannel?	Multiple links bundled as one logical link via LACP/PAgP/static; needs matching settings.
Difference between OSPF and BGP?	OSPF: interior link-state, fast, uses areas and multicast. BGP: exterior path-vector over TCP 179, policy-based between AS.
OSPF DR/BDR?	Reduce adjacencies on multi-access segments; elected by priority then router ID.
Why OSPF neighbour stuck in ExStart?	Usually MTU mismatch or duplicate router ID.
BGP Active state?	TCP session is not establishing; check reachability, port 179, configuration.
Administrative distance values?	Connected 0, static 1, eBGP 20, EIGRP 90, OSPF 110, RIP 120, iBGP 200.
TCP vs UDP?	TCP reliable connection-oriented; UDP connectionless, low overhead.
Explain 3-way handshake.	SYN, SYN-ACK, ACK; establishes sequence numbers and options.
What does TCP RST mean?	Abort/refuse the connection: closed port, firewall action, or application reset.
What is ARP and what is gratuitous ARP?	IP-to-MAC resolution; gratuitous ARP announces own mapping, used for failover and duplicate detection.
What is TTL?	Hop limit; decremented per router; prevents loops; basis of traceroute.
What is DHCP DORA and relay?	Discover, Offer, Request, ACK; relay forwards broadcast to server in another subnet.
What is NAT/PAT?	Translating private to public addresses; PAT also changes source port to share one IP.
Stateful vs stateless firewall?	Stateful tracks sessions and permits return traffic automatically; stateless filters each packet independently.
What is App-ID?	Palo Alto identifies applications regardless of port, using signatures, decoding and heuristics.
Zones on Palo Alto?	Logical groups of interfaces; policy is applied between zones.
Pre-NAT vs post-NAT in policy?	Policy uses original IPs and post-NAT destination zone.
IPSec Phase 1 and 2?	Phase 1 builds IKE SA to authenticate and protect negotiation; Phase 2 builds IPSec SAs to protect data.
Panorama templates vs device groups?	Templates: network/device config. Device groups: policies and objects.
How do you check why traffic is denied?	Traffic log, rule name and session end reason, test security-policy-match, check zones and App-ID.
L1 SOC alert handling?	Validate alert, check context/logs, determine true or false positive, escalate with evidence, document in the ticket.
When do you escalate?	When SLA risk, scope or impact exceeds L1 authority, no resolution via runbook, or critical systems/security incident.
What is the difference between incident and problem?	Incident: service disruption to restore. Problem: root cause of recurring incidents.

Behavioural prompts: have a 60-second answer ready

- **Tell me about yourself:** Network Security Engineer with about 1.4 years in Palo Alto TAC, supporting enterprise NGFW, Panorama, IPSec/SSL VPN and GlobalProtect issues with Wireshark and RCA reports; strong in routing and switching fundamentals; studying CCNA; looking to grow in a Network/SOC operations role.
- **A tough case you solved (STAR):** choose a VPN or packet-capture case: Situation, Task, the steps you took, the root cause, the result and what you documented.
- **24x7 shifts:** comfortable with rotational shifts and handover discipline.
- **Weak areas (SDA/SIEM):** be honest: 'I have not worked on SDA/SIEM hands-on, my background is firewall, routing and switching, and I learn quickly; I have started covering the concepts.'
- **Questions to ask:** tools used (monitoring, ticketing, SIEM), shift pattern, escalation matrix, training for SDA.

14. Last-day revision plan and cheat sheet

Tonight and tomorrow morning

Time	Do
Tonight (60-90 min)	Read chapters 1, 4, 2, 7 and say the answers aloud. Draw the TCP handshake, ARP flow and STP election from memory.
Tonight (30 min)	Chapter 8: OSPF states and stuck causes, BGP states, HSRP. Practise 5 subnetting questions.
Morning (45 min)	Chapter 9 firewall flow and VPN troubleshooting, chapter 12 scenarios S1, S3, S4, S6, S8.
Before the interview	Rehearse 'tell me about yourself', one STAR incident, and honest answers for SDA/SIEM. Sleep well.

One-page cheat sheet

Topic	Remember
OSI	7 App, 6 Pres, 5 Sess, 4 Transport (segment), 3 Network (packet), 2 Data link (frame), 1 Physical (bits)
Per hop	MAC changes, IP same, TTL-1
Ports	22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 67/68 DHCP, 80, 443, 123 NTP, 161/162 SNMP, 179 BGP, 389 LDAP, 500/4500 IKE, 514 syslog, 3389 RDP, 1985 HSRP
TCP	SYN, SYN-ACK, ACK. FIN/ACK ×2. RST abort. 3 dup ACK = fast retransmit
STP	Lowest bridge ID root; cost 1G=4; RSTP 1-2 s; PortFast + BPDU Guard on edge
VLAN	802.1Q tag 4 bytes; native untagged; SVI up needs active port
OSPF	Proto 89, 224.0.0.5/6, hello 10 / dead 40, DR by priority then RID, stuck ExStart = MTU
BGP	TCP 179, keepalive 60 / hold 180, Active = TCP failing, best path: Weight, LP, AS-path, MED
HSRP	Priority 100, preempt off by default, hello 3 / hold 10, UDP 1985
Firewall	Zones, first-match policy, pre-NAT IP + post-NAT zone, session table, App-ID
IPSec	P1 UDP 500/4500, P2 ESP, proxy-ID mismatch = P2 fail, NO_PROPOSAL_CHOSEN = P1 mismatch
Wireshark	SYN without reply = drop; SYN then RST = refused; Dup ACK/retrans = loss; ZeroWindow = slow receiver

FINAL NOTE

You do not need to know everything. Show a clear method, be honest about what you have not used, and use your real Palo Alto TAC examples. That is what a Network/SOC L1 interviewer is looking for. Good luck tomorrow.